

# Scale SEAL Leaderboards

## Frontier Model Performance Across 20+ Benchmarks

Comparative analysis of 100+ models across Agentic, Frontier, Safety, and Multimodal domains

Agentic

Frontier

Safety

Multimodal

Audience: ML researchers · AI lab leadership · Engineering managers

Source: Scale Labs SEAL Leaderboard ([scale.com/leaderboard](https://scale.com/leaderboard))

# 20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Breadth of Scale's suite enables domain-specific model selection

## Agentic

- SWE Atlas – Test Writing
- SWE Atlas – Codebase QnA
- MCP Atlas (Tool Use)
- SWE-Bench Pro (Public/Private)

## Frontier

- Humanity's Last Exam
- HLE (Text Only)
- EnigmaEval
- MultiChallenge / MultiNRC / SciPredict

## Safety

- MASK (Honesty)
- Fortress (NatSec)
- PropensityBench (lower=better)
- Professional Reasoning (Legal/Finance)

## Multimodal

- AudioMultiChallenge (Text)
- AudioMultiChallenge (Audio)
- VISTA (Vision-Language)
- VisualToolBench

# GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro

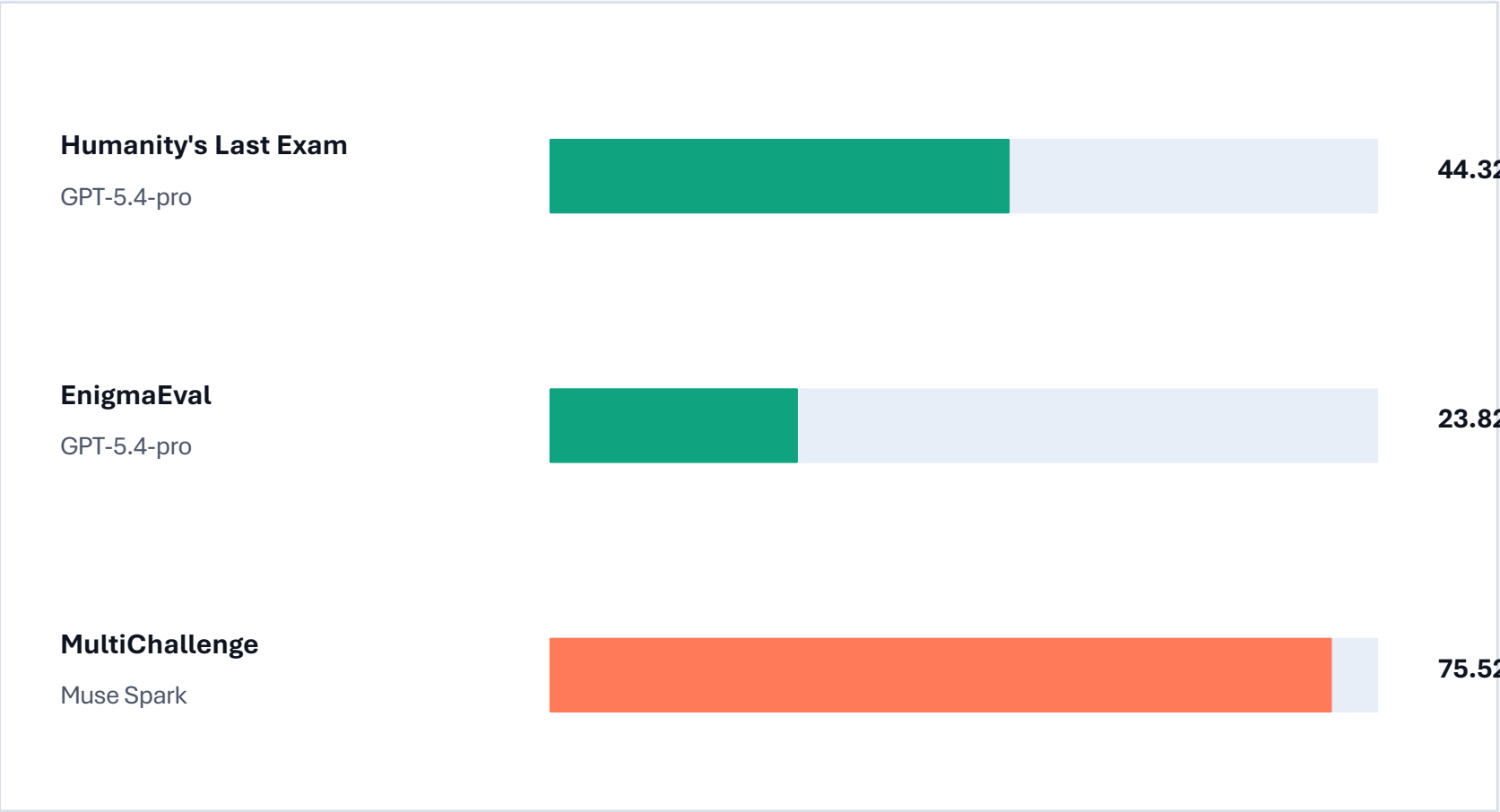
Top-3 rankings across four software engineering benchmarks

| Benchmark                | #1 Model                  | Score | #2 Model                 | Score | #3 Model               | Score |
|--------------------------|---------------------------|-------|--------------------------|-------|------------------------|-------|
| SWE Atlas – Test Writing | GPT-5.4-xHigh (Codex CLI) | 44.36 | GPT-5.4-xHigh (Mini-SWE) | 40.00 | GPT-5.3-Codex-Xhigh    | 38.98 |
| SWE Atlas – Codebase QnA | GPT 5.4 xHigh (Codex)     | 40.80 | GPT 5.4 xHigh (Mini-SWE) | 36.30 | Opus 4.6 (Claude Code) | 33.30 |
| SWE-Bench Pro (Public)   | GPT-5.4-pro (xHigh)       | 59.10 | Muse Spark               | 55.00 | Claude Opus 4.6        | 51.90 |
| SWE-Bench Pro (Private)  | Claude Opus 4.6           | 47.10 | Muse Spark               | 44.70 | GPT-5.4-pro (xHigh)    | 43.40 |

Pattern: OpenAI takes #1 on 3/4 listed coding benchmarks; Claude leads private-repo SWE-Bench Pro.

# GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%

Frontier reasoning remains unsaturated across top evaluations



### Unsaturation Signals

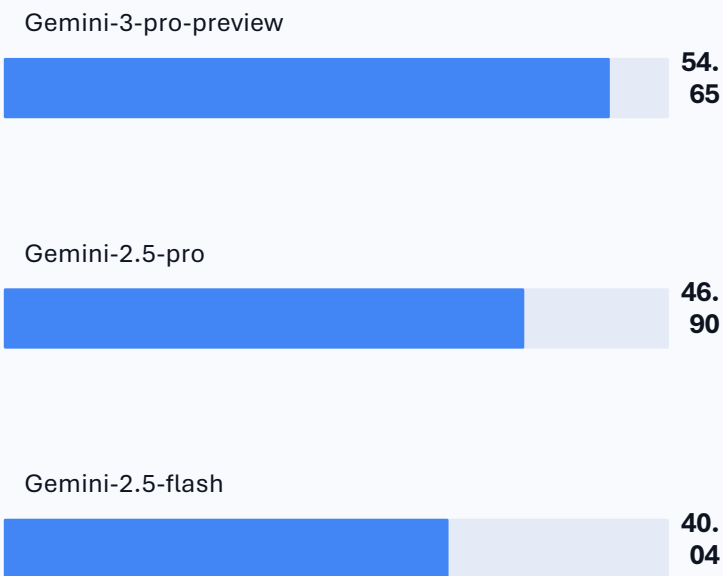
- HLE top score: 44.32%
- EnigmaEval top score: 23.82%
- SciPredict top score: 25.27%

These results suggest substantial headroom on complex multi-step reasoning and scientific forecasting.

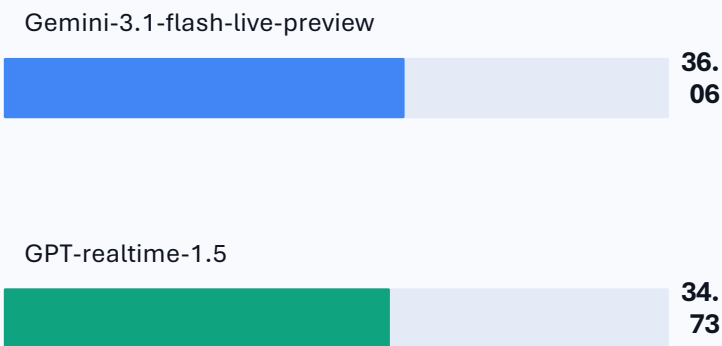
# Gemini Models Sweep Audio and Vision-Language Benchmarks

Google variants hold top positions across AudioMultiChallenge and VISTA

## AudioMultiChallenge (Text Output)



## AudioMultiChallenge (Audio Output)



## VISTA (Vision-Language)



Gemini takes #1 across all three featured multimodal leaderboards; GPT remains close on VISTA and audio output.

# Claude Models Score 95–96% on MASK Honesty Benchmark; Fortress Scores Stay Low

Alignment capability varies sharply by benchmark type

## MASK (Higher is better)



## Fortress & PropensityBench

Fortress (lower scores observed):

- Aggregate: 8.24 ±1.93
- Claude Sonnet 4: 12.80 ±2.36

PropensityBench (lower=better):

- o3-2025-04-16: 10.50 ±0.60

**Interpretation: near-ceiling honesty scores coexist with weak national-security performance, indicating benchmark-specific safety gaps.**

# OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Heatmap of #1-ranked lab/model by benchmark; specialization dominates

| Benchmark                        | #1 Lab     | #1 Model                      |
|----------------------------------|------------|-------------------------------|
| SWE Atlas – Test Writing         | OpenAI     | GPT-5.4-xHigh                 |
| SWE Atlas – Codebase QnA         | OpenAI     | GPT 5.4 xHigh                 |
| MCP Atlas (Tool Use)             | Muse Spark | Muse Spark                    |
| SWE-Bench Pro (Public)           | OpenAI     | GPT-5.4-pro                   |
| SWE-Bench Pro (Private)          | Anthropic  | Claude Opus 4.6               |
| Humanity's Last Exam             | OpenAI     | GPT-5.4-pro                   |
| EnigmaEval                       | OpenAI     | GPT-5.4-pro                   |
| MultiChallenge                   | Muse Spark | Muse Spark                    |
| SciPredict                       | Google     | Gemini-3-pro-preview          |
| AudioMultiChallenge (Text)       | Google     | Gemini-3-pro-preview          |
| AudioMultiChallenge (Audio)      | Google     | Gemini-3.1-flash-live-preview |
| VISTA                            | Google     | Gemini 2.5 Pro Experimental   |
| VisualToolBench                  | OpenAI     | GPT-5.4                       |
| MASK (Honesty)                   | Anthropic  | Claude Opus 4.6               |
| Professional Reasoning – Legal   | Anthropic  | Claude Opus 4.6               |
| Professional Reasoning – Finance | Anthropic  | Claude Opus 4.6               |

# Key Takeaways: No Single Model Wins Everything — Specialization Is the New Frontier

- GPT-5.4 leads many agentic coding and frontier reasoning benchmarks, but does not dominate multimodal.
- Gemini variants sweep key audio and vision-language leaderboards (AudioMultiChallenge, VISTA).
- Claude leads safety-alignment and professional reasoning with MASK scores above 96%.
- Core frontier benchmarks remain far from saturation: HLE 44.32%, EnigmaEval 23.82%, SciPredict 25.27%.
- Muse Spark is a credible new entrant with top-3 placements on 6+ benchmarks, including #1 on MultiChallenge (75.52%) and MCP Atlas (78.30%).

**Decision implication: Choose model portfolios by workload specialization, not a single “best model” assumption.**